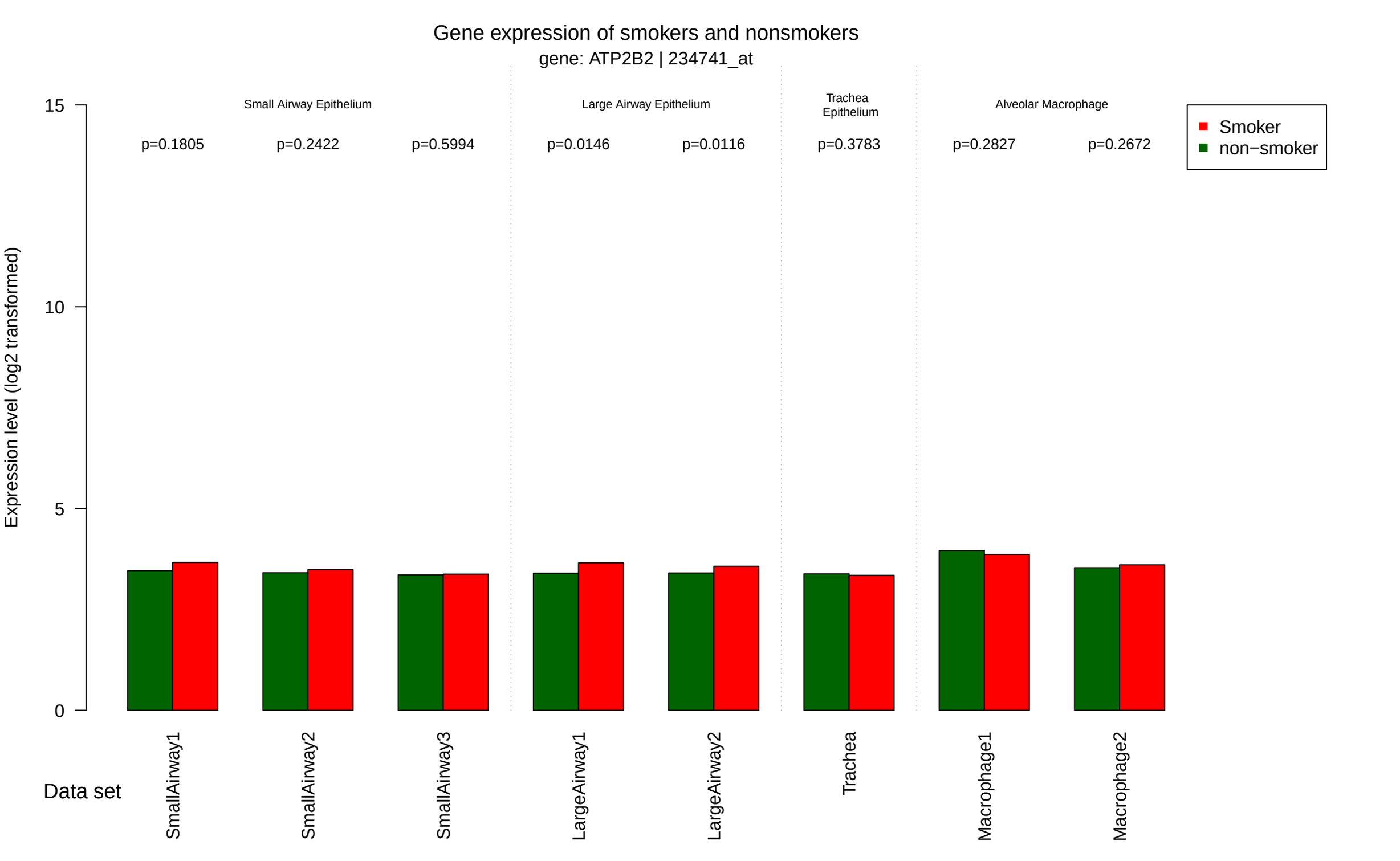
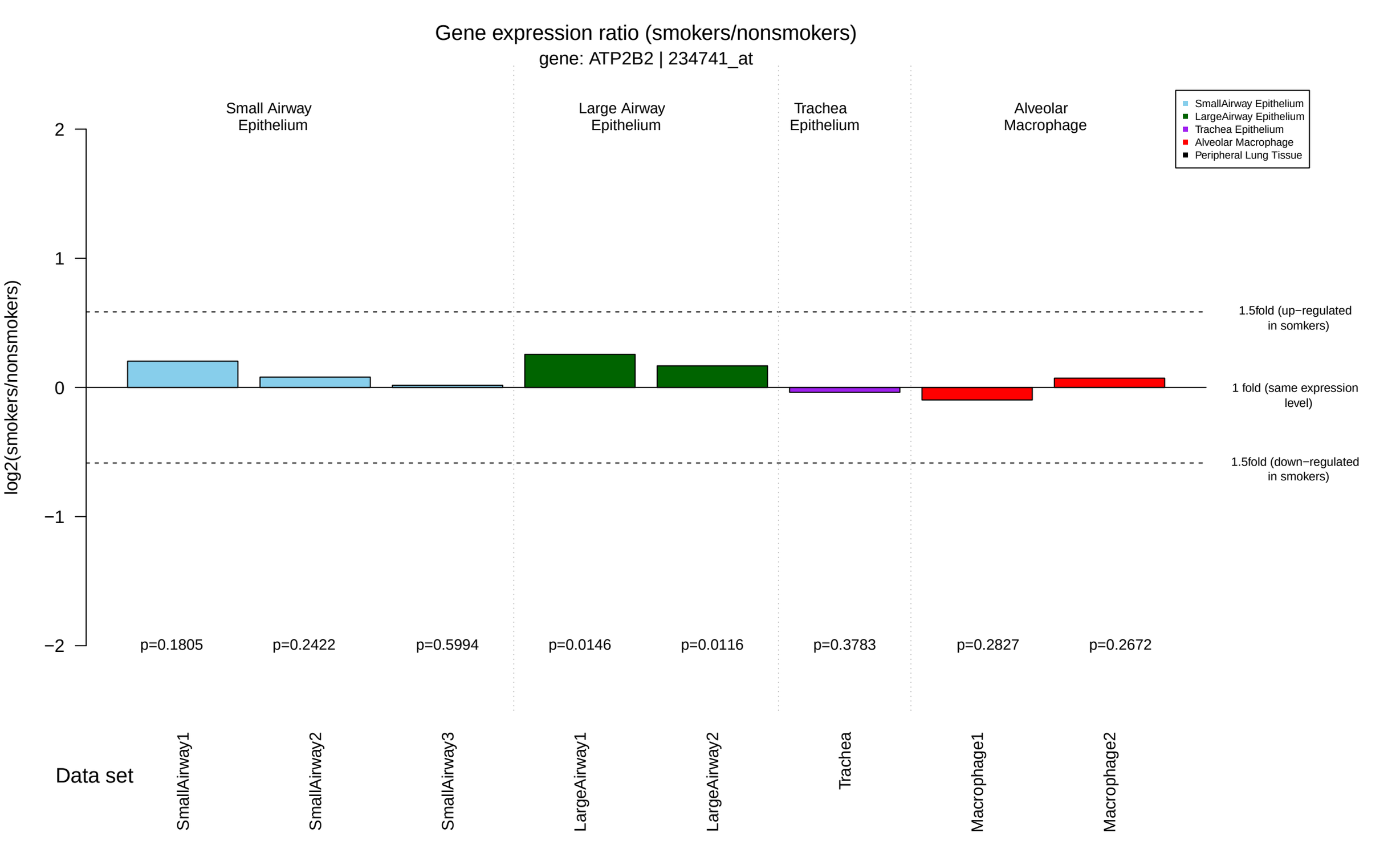
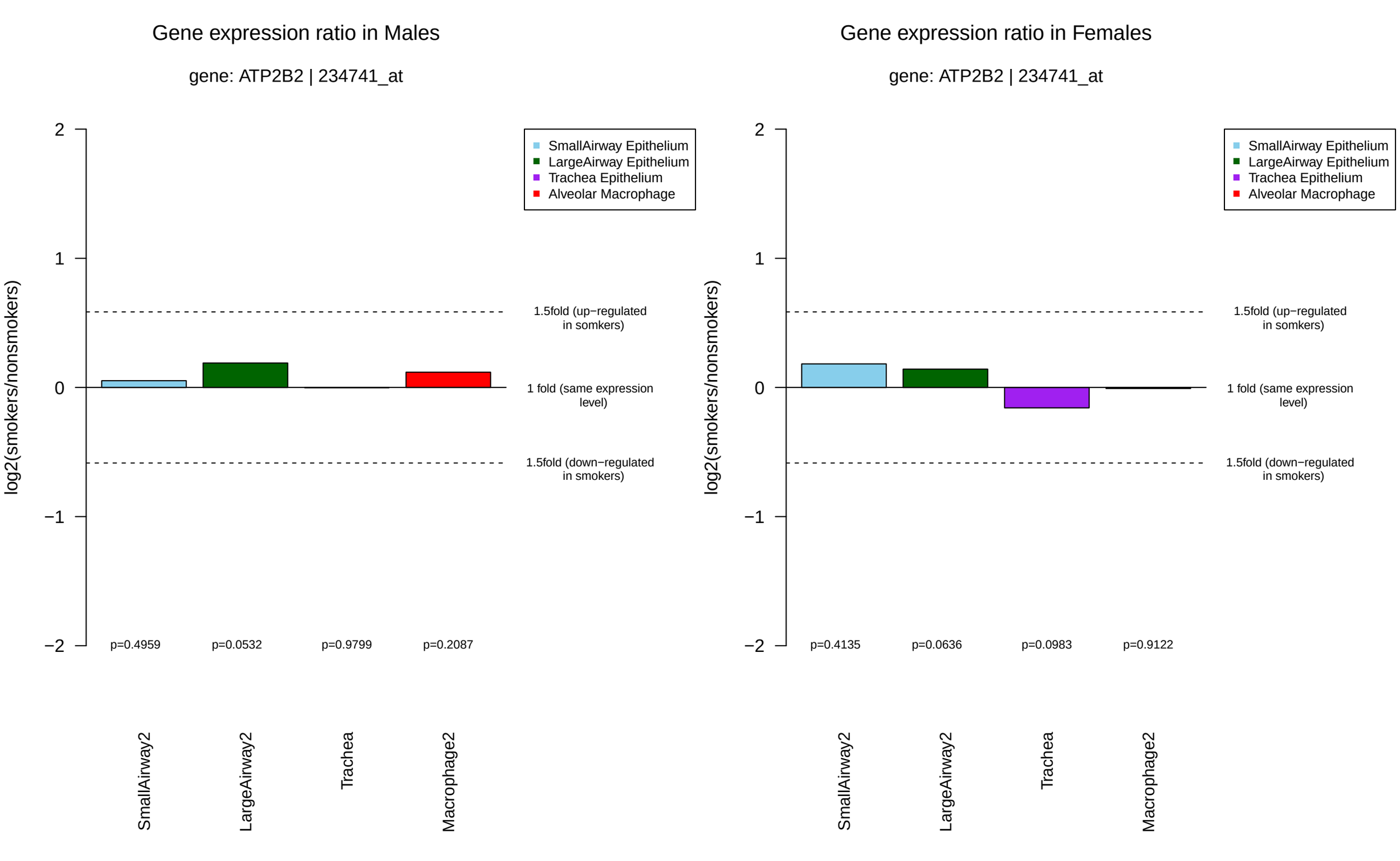
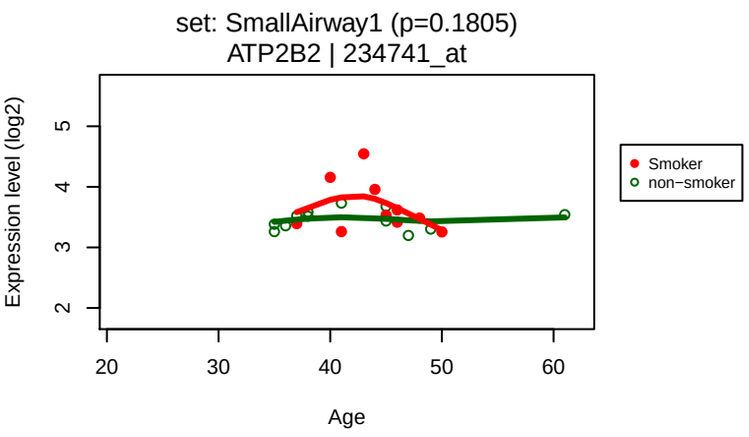
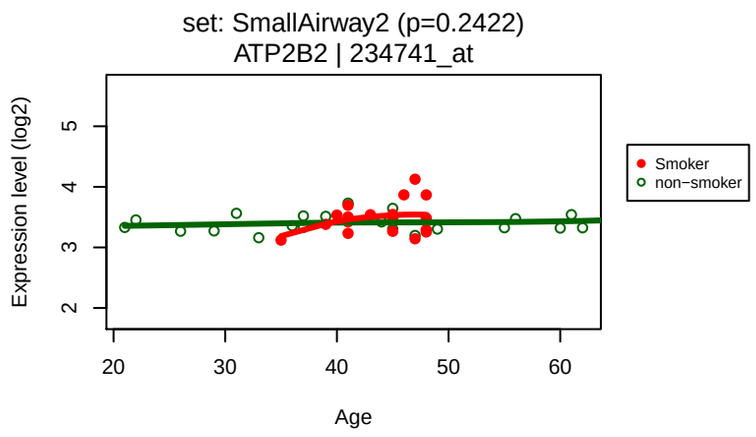
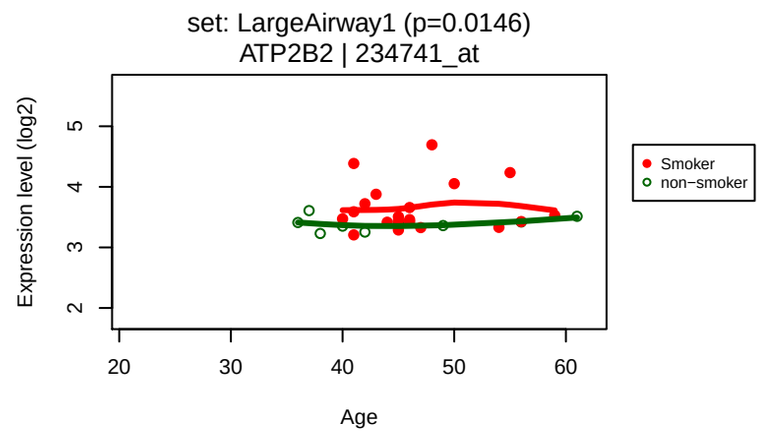

Gene expression of smokers and nonsmokers
gene: ATP2B2 | 234741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1805
p=0.2422
p=0.5994
p=0.0146
p=0.0116
p=0.3783
p=0.2827
p=0.2672
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

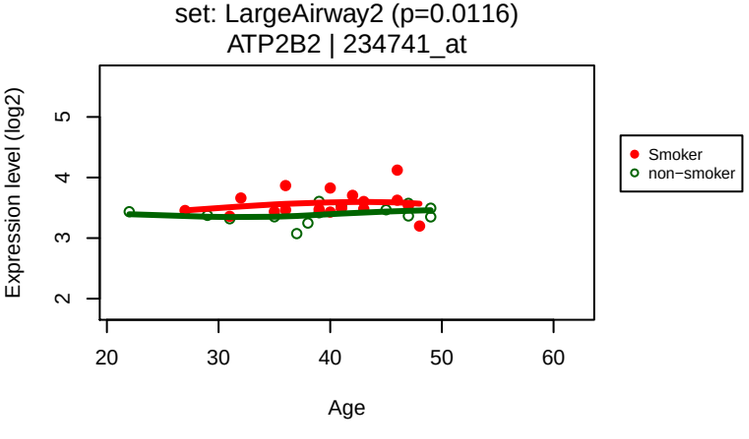

Gene expression ratio (smokers/nonsmokers)
gene: ATP2B2 | 234741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1805
p=0.2422
p=0.5994
p=0.0146
p=0.0116
p=0.3783
p=0.2827
p=0.2672
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

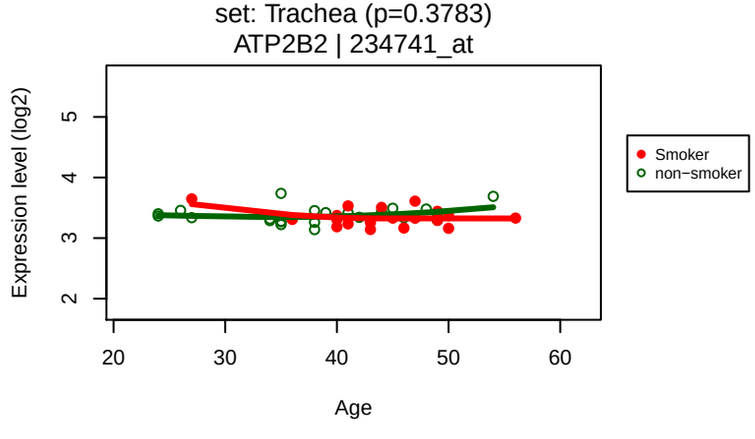

Gene expression ratio in Males
gene: ATP2B2 | 234741_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4959
p=0.0532
p=0.9799
p=0.2087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2B2 | 234741_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4135
p=0.0636
p=0.0983
p=0.9122
SmallAirway2
LargeAirway2
Trachea
Macrophage2

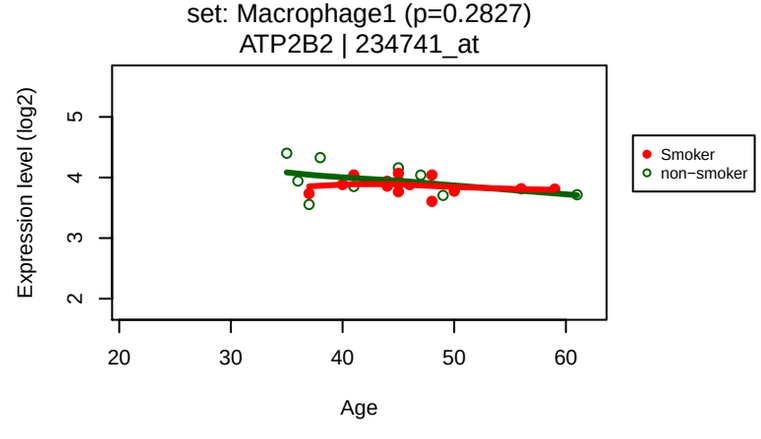

set: SmallAirway1 (p=0.1805)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker

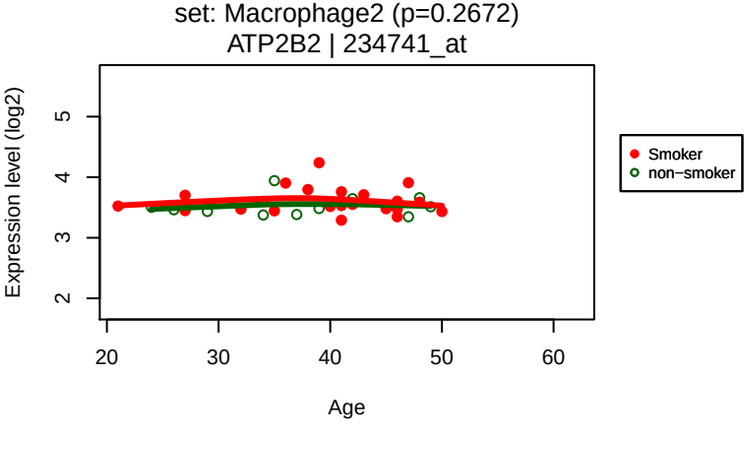

set: SmallAirway2 (p=0.2422)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker

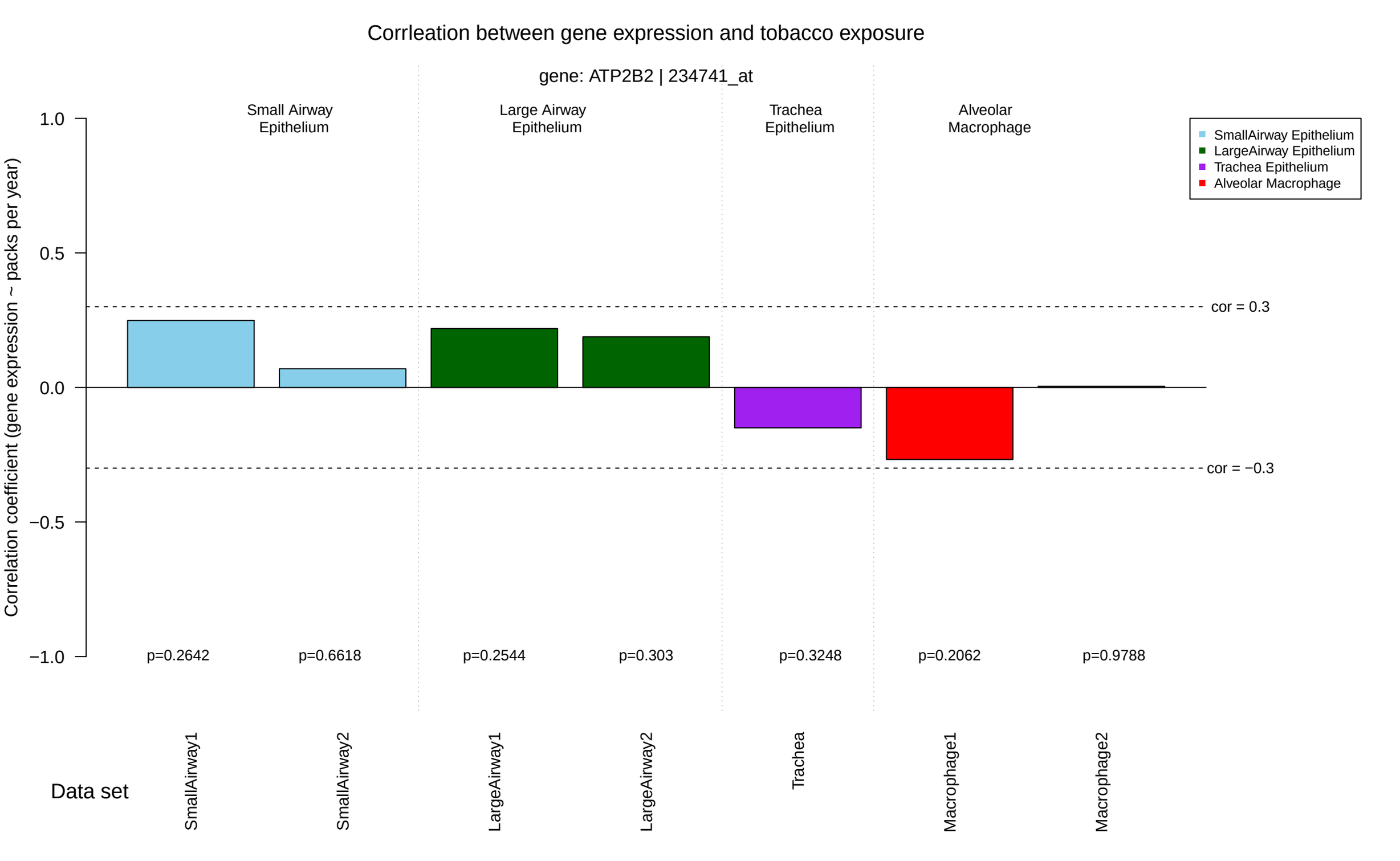

set: LargeAirway1 (p=0.0146)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0116)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3783)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2827)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2672)
ATP2B2 | 234741_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2B2 | 234741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2642
p=0.6618
p=0.2544
p=0.303
p=0.3248
p=0.2062
p=0.9788
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2